

BANJOU TAHARA

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5-36 Shikinaicho, Izumiotsu-shi, Osaka 595-0035, Japan

EDUCATION

The University of Osaka, Japan

April 2026 – Present

Master's Program

Department of Systems Innovation, Graduate School of Engineering Science

Enrolled in the Department.

The University of Osaka, Japan

April 2022 – March 2026

Bachelor of Engineering

Department of Systems Science, School of Engineering Science

Graduated First in the Department (Valedictorian).

CAREER OBJECTIVE

To advance next-generation spatial computing by shifting the paradigm from rigid physical limitations and isolated VR/AR environments to a ubiquitous Real-Cyber co-existence. Leveraging a rigorous multidisciplinary background, I aim to synergize computational imaging, the psychophysics of material appearance, and HCI. My ultimate goal is to dissolve the boundaries between digital and physical domains, bringing the boundless flexibility of Cyber-Worlds into the real physical world to elevate human quality of life (QoL).

PROJECTS

Spatial Overwriting via Volumetric Scanning Projection Mapping

2025 - Present

Developed a novel Projection Mapping (PM) framework to realistically manipulate physical material appearance (color, gloss, and texture) without permanent surface modification. Addressed the fundamental limitation of flat-looking projections by introducing a volume-scanning approach to reproduce "mesoscopic" 3D micro-structures (e.g., fine bumps and roughness).

Proposed Method & Results: Designed a volumetric scanning projection system that applies rapid, microscopic 2D orbital vibrations to the projection target. By synchronizing these movements with the high-speed switching of normal-mapped depth layer images, the system successfully enables the perception of textures that bring out 3D bump geometries within the temporal integration window of the human visual system.

Future Vision: To address current technical challenges, immediate future work focuses on refining hardware precision to eliminate motion blur caused by the induced vibrations. Following this optimization, I plan to conduct user studies and rigorous psychophysical evaluations to investigate the relationship between the 3D displacement geometry data of textures and human material perception.

PUBLICATIONS AND PRESENTATIONS

Domestic Conference / Non-Peer-Reviewed

1. Banjou Tahara, Daisuke Iwai, "Geometric Relief Representation via High-Speed Volumetric Projection synchronized with 2D Micro-Vibration", The 70th Annual Conference of the Institute of Systems, Control and Information Engineers (SCI'26), Japan, pp.772-778, March.24- March.26, 2026.

ACADEMIC AWARDS

Kusumoto Award (Valedictorian / Top Graduating Student)

March 2026

Awarded by The University of Osaka to the top graduating student in the Department of Systems Science, School of Engineering Science, for outstanding academic excellence.

SICE Outstanding Student Award

March 2026

Conferred by The Society of Instrument and Control Engineers (SICE) to recognize students who demonstrate exceptional academic performance and exemplary character.

TECHNICAL STRENGTHS

Core Disciplines	Computer Vision, Computer Graphics, Optics, Visual Psychology, Signal Processing, Measurement & Control Engineering, Human-Computer Interaction, Applied Physics, Electronic Engineering, System Optimization.
Graphics & Vision	OpenCV, OpenGL, Physically Based Rendering (PBR), Projector-Camera Systems, Projection-Mapping, Material Appearance Science.
AI & Robotics	PyTorch, Multimodal World Models for Robot Learning, Generative AI, Image Tokenizer Training.
Sensor Fusion	Multi-Sensor Integration (LiDAR, IMU, GNSS) for Autonomous Driving Development.

INTERN EXPERIENCE

Matsuo Institute, Inc.

Research Engineer (Intern)

May 2025 - April 2026

Tokyo, Japan

Engaged in cutting-edge research and development on **Multimodal World Models** for Robot Learning.
Developed and trained custom image tokenizers using PyTorch to build robust neural network baselines for robotic decision-making.
Implemented large-scale machine learning pipelines to synthesize multimodal sensor data, bridging generative AI with physical robotic control.

KG Motors Co., Ltd.

Autonomous Driving Software Engineer (Intern)

June 2024 - December 2025

Hiroshima, Japan

Developed an end-to-end **Autonomous Driving System** deployed on a physical vehicle through the integration and synchronization of heterogeneous sensors (LiDAR, IMU, and GNSS).
Implemented a high-fidelity **Digital Twin** environment synchronized with the physical system to simulate and validate vehicle kinematics, sensor behaviors, and path-planning logic.

EXTRA-CURRICULAR & COMPETITIONS

Autonomous Driving AI Challenge 2025 (JSAE)

July 2025 - October 2025

Competed in an autonomous driving lap time racing competition organized by the Society of Automotive Engineers of Japan (JSAE).

Achieved **4th place in the Qualifying Round** and **7th place in the Final Round** out of 249 participating teams.

Developed and optimized path-planning algorithms and vehicle control logic to minimize lap times while ensuring safe obstacle avoidance.

PERSONAL TRAITS

Growth-Oriented Research Mindset (Positive Thinking): Views technical setbacks and experimental failures not as roadblocks, but as essential data points and catalyst for growth. Applies a resilient, constructive mindset to overcome the inherent challenges of pioneering research.

Synergistic Hardware-Software Intuition: Possesses a rare, hands-on understanding of the synergy between hardware constraints and software optimization. Experienced in bridging the gap between digital algorithms (computer vision, ML, digital twins) and physical mechanics (projection systems, real-vehicle heterogeneous sensors).